

JDScience 11+ Non-Verbal Reasoning Practice Paper 1 — Answers and Explanations

Non-Verbal Reasoning · Answers and Explanations

Name _____

Date _____

Score _____

Recommended time: —

Questions: 40

Calculator: Not required unless a question says otherwise

Instructions

1. Issue separately from the pupil paper.

This is an original JDScience practice paper written for school entrance exam preparation. It is not an official GL Assessment, CEM, CSSE, school or publisher paper.

Mark scheme

1. **A** — Each figure adds one filled dot. After four dots, five dots come next.
2. **C** — One square is added each time. Five squares follow four.
3. **B** — The arrow turns 90° clockwise each step. After pointing up, it points right again.
4. **E** — One horizontal line is added each time. Five lines come next.
5. **D** — One nested square is added each step. Five nested squares follow.
6. **B** — The hand moves a quarter-turn clockwise each time ($12 \rightarrow 3 \rightarrow 6 \rightarrow 9 \rightarrow 12$).
7. **A** — One vertical bar is added each time. Five bars come next.
8. **C** — Shading increases then restarts empty: empty $\rightarrow$ quarter $\rightarrow$ half $\rightarrow$ full $\rightarrow$ empty again.
9. **E** — A 90° clockwise turn points the triangle to the right.
10. **D** — Rotating the upright L (L) 90° clockwise gives ┐ .

Mark scheme

11. B — A 180° turn moves the top-left dot to the bottom-right corner.

12. A — After 0° , 90° and 180° , the next clockwise quarter-turn is 270° .

13. C — From Γ , a 90° anti-clockwise turn returns the L to $\perp$.

14. E — A 180° turn moves left shading to the right side.

15. D — Next quarter-turn after 180° is 270° .

16. B — A vertical mirror turns $\perp$ into $\lrcorner$.

17. A — Flipping top to bottom moves the top-left dot to bottom-left.

18. C — A vertical mirror swaps left and right: top-right becomes top-left.

19. E — Left shading becomes right shading under a vertical mirror.

20. D — A left-right mirror turns a right-pointing arrow into a left-pointing arrow.

Mark scheme

21. A — Top shading becomes bottom shading under a horizontal mirror.

22. B — Pattern pairs outer square+circle, outer circle+square, outer square+dot, so outer circle+dot completes the set.

23. C — Reading left-to-right, top-to-bottom: 1, 2, 3, then 4 bars.

24. D — Clockwise corners: TL → TR → BR → BL.

25. E — Each row is n , $n+1$, $n+2$. Bottom row: 3, 4, 5.

26. A — Top: circle → square, so bottom: square → circle.

27. B — Shading position cycles: left, right, top, bottom.

28. C — Same additive row pattern: bottom row ends with 5 lines.

29. D — A, B, D and E are circles; C is a square.

30. E — A–D are arrows; E is a plus sign with no arrowhead.

Mark scheme

31. A — Four figures have three dots; D has four.

32. B — A–D are half-shaded squares; E is a half-shaded circle.

33. C — A–D are rotations of the same L-shape; E is nested squares.

34. D — In A–D the dot sits on the ring's edge; in E the dot is in the centre.

35. E — Most show circle then square left-to-right; D reverses the order.

36. B — A left–right fold mirrors the punch: hole at top-left also appears at top-right.

37. C — Reversing left–right order: triangle–circle becomes circle–triangle.

38. D — When that panel becomes the front face, a single black dot remains visible on the front square.

39. E — A vertical flip moves left shading to the right side.

40. A — Moving the leftmost circle to the far right leaves square, triangle, circle.